Current Solved Paper BT – 601 (Virology)… 2023

1. **What is a reason behind the spontaneous mutations? (2)**

Spontaneous Mutation that arise naturally during replication

REASON

- Mechanism of genome replication
- RNA dependent RNA polymerase shows higher error rate than DNA dependent DNA polymerase

2. **What is the mode of transmission of arena viruses? (2)**

Arena viruses are a family of viruses that are usually transmitted from **rodents to humans.**

Arena viruses are shed into the environment in the urine, saliva, or droppings of infected rodent hosts. **People become infected by breathing in the virus after rodent urine, droppings, or nesting materials are stirred up, such as during cleaning.**

3. **Describe the host range virus? (3)**

- Host ranger refers to the spectrum of host cells in which virus can multiply.
- Most viruses infect only specific types of cells in one host species.
- Host range is determined by the specific attachment site on the host cells surface and the availability of host cellular factors.
- Viruses that infect bacteria are called bacteriophages, or phages.

4. **What are the 3 most efficient theories for evolution? (3)**

FIRST THEORY: They may have originated and evolved with the other life forms

SECOND THEORY: they may have arisen from segments of the cellular nucleic acids which acquired the ability to replicate at the expense of host cell.

THIRD THEORY it is Regressive theory. According to this theory, Viruses arose from free living organisms which gradually lost genetic information. Until they became totally dependent on the biosynthetic pathways of their host cells

5. **What are the functions of different protein of paramyxoviridae? (5)**

PROTEINS

N – The nucleocapsid protein associates with genomic RNA and protects the RNA from nuclease digestion

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P – The phosphoprotein binds to the N and L proteins and forms part of the RNA polymerase complex

M – The matrix protein assembles between the envelope and the nucleocapsid core; it organizes and maintains virion structure

F – the fusion protein projects from the envelope surface as a trimer, and mediates entry

H/HN/G – the cell attachment proteins span the viral envelope and project from the surface as spikes.

L – The large protein is the catalytic subunit of RNA dependant RNA polymerase (RDRP)

Accessory proteins – a mechanism known as RNA editing allows multiple proteins to be produced from the P gene.

6. **Phase contrast microscope principles and it applications? (5)**

Phase-contrast microscopy (PCM) is an optical microscopy technique that converts phase shifts in light passing through a transparent specimen to brightness changes in the image. Phase shifts themselves are invisible, but become visible when shown as brightness variations.

Principle:

When light passes through cells, small phase shifts occur. These phase shifts are converted into changes in amplitude, which can be observed as differences in image contrast.

Application:

Cell culture

Live cell imaging

Living cells can be observed in their natural state without previous fixation or labeling.

7. **Genome structure of retroviruses? (5)**

Genome of retrovirus:

Single stranded RNA viruses

Positive sense

Enveloped

Changes to DNA by RT enzyme

There are three important genes

1. GAG

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2. ENV
3. POL

8. Structure of retroviruses:

It is the single stranded positive sense of RNA viruses. It is icosahedral in shape. All the retrovirus is surrounded by viral envelope. It contains two identical RNA and reverse transcriptase enzymes.

9. Latent virus types? (3)

LATENT VIRAL INFECTIONS

Chronic persistent infections

Latent Occult Viral infections

10. Differential interference contrast microscopy:

Also known as Nomarski interference contrast microscopy

used to enhance the contrast in unstained, transparent samples

DIC works on the principle of interferometry to gain information about the optical path length.

Application:

DIC is a more expensive, label-free microscopy technique with a high sensitivity to thin cellular material, even when it is located within thick tissue. It is **useful for rendering contrast in transparent samples and gives brilliant pseudo-3D relief shading images**.

11. Replication Toga virus genome?

--Viral RNA acts as both genome as well as m RNA

--After fusion of viral capsid with cell receptor

--RNA released in cytoplasm

12. Viral classification of picornavirus?

- A picornavirus is a virus belonging to the family Picornaviridae, and order Picornavirales
- Vertebrates, including humans, serve as natural hosts. Birds also serve as host
- Picornaviruses are nonenveloped viruses that represent a large family of small, cytoplasmic, plus-strand RNA (~7.5kb) viruses
- 30-nm icosahedral capsid

13. Name the 10 positive sense RNA Plant virus?

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BROMOVIRIDAE

• Cucumovirus

• Bromovirus

• Ilavirus

• Alfamovirus

• Ourmiavirus

• Tobamovirus

• Hordievirus

• Pecluvirus

• Furovirus

• Pomovirus

• Bemivirus.

14. Principle of Flourescence microscope:

The specimen is illuminated with light of a specific wavelength which is absorbed by the fluorophores, causing them to emit light of longer wavelengths • Florescent dyes are used for specimen staining • Examples of Flourescent stains: DAPI and Hoechst

15. What is neutralization test?

- It is a serological test that detects the presence of viral neutralizing antibodies.
- The antibodies bind to viral particles.
- These antibodies block viral infectivity and prevent from viral infections.

16. Baltimore classification? (5)

Baltimore system of classification 1971

• Type of nucleic acid

• Mode of replication

• Host organism

• Diseases caused

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17. Characteristics of virus? (5)

- Following are the general properties of viruses
- Non living structure
- Non cellular
- Contain a protein coat known as capsid
- Contain the nucleic acid core containing DNA or RNA
- Can produce in host cells
- Some virus may have spikes to help attach to the host cells

..Viruses Words originate.....? It derives from the **Latin word virus meaning toxin or poison**

…Membrane fusion is the mechanism of viral what in host cell... Endocytosis of the virus into the host cell

…Size of mouth and foot….. 27nm

…Virus that infect the bacteria.?..... Bacteriophages

…Smallest viruses?..... **Adenovirus is the smallest virus**

…Equine encephalomylatic kill.....24hours

…Rhabdovirus in which Rhabdo means?.. rod

---contril Mosquito bone viruses …… Aedes Egyptei

---The Protein sheel of Viruses???.... Made up of protein sub-units (protomers)

---Which microscope is used for transparent sample.?... DIC MICROSCOPY • Also known as Nomarski interference contrast microscopy

---Father of microbiology is ….Antony wan Leuwenhoek

---First discovered term vaccine is …Edwar jenner

---Who discovered first cell line is ,….carrel.

---Immunodiffusion is performed using _____1 % agarose

---Tissue sample stored at degree for years…-80*C

---Toroviruse belong to a family is …Coronaviridae.

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---Virus infection cell release ….interferon.

----Smallest viruse name… Picorona virus.

---Misfiled of protein is known as ….prions.

---Mad cow disease is caused by……. prions.

---Mutation is _____ in HIV. .Very high

---Viruses store their genetic information into ____types of genome…. Six

---Nucleic acid of virus is ___ ….Capsid

---Live vaccines are used to protect against Rubella,, …….Rotavirus

---All virus family contain the suffix…… viridae.

---Creutzfeldt-Jakob……. disease caused by prions.

----An icosahedral ….is a polygon with 12 vertices (corner), and 30 edges.

---A vaccine……. is a biological preparation that provides active acquired immunity to a particular disease.

----The host cell takes in the viral particle through the process of …..endocytosis

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